

Algorithms Problem Solving

Level 3

Check Sparse Matrix



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Definition

Sparse Matrix Detection

 Program determines whether a matrix contains enough zero elements to satisfy the chosen sparsity rule.

 In this implementation, the matrix is considered sparse when the number of zeros is at least half of its total elements.

Explanation

Zero Counting

-  ``CountNumberInMatrix()`` is reused with ``0`` as the target value.

-  The returned count represents the number of zero elements.

Matrix Size Calculation

-  Total matrix capacity is calculated using ``rows * cols``.

-  This value represents the total number of positions that may contain zero or non-zero values.

Sparsity Threshold

-  The implementation compares the zero count with half of the matrix size.

-  Using ``ceil(size / 2.0)`` avoids losing the fractional half when the total number of elements is odd.

Boolean Decision

-  If the zero count satisfies the selected threshold, the function returns ``true``.

-  Otherwise, it returns ``false``.

Procedures

1. Define `CountNumberInMatrix()`.
2. Count all zero elements.
3. Calculate `MatrixSize = rows * cols`.
4. Calculate the zero threshold.
5. Return `true` when the zero count reaches the threshold.
6. Display whether the matrix is sparse.

💡 Example

matrix1 = { {0,0,12}, {0,0,1}, {0,0,9} };

➡ Result

Yes: It is Sparse

Note

- ◆ "Sparse matrix" is a broader mathematical/computational concept and different applications may use different practical density thresholds.
- ◆ This specific exercise defines sparse as zero elements being at least half of the matrix.
- ◆ Sparse data can benefit from specialized storage formats such as *CSR* or *CSC* when matrices become large.

⊘ Warning

- ◆ Integer division such as ``MatrixSize / 2`` can lose the fractional part for odd matrix sizes.
- ◆ Do not assume that every sparse-matrix definition uses exactly the same threshold.
- ◆ A matrix being sparse does not automatically mean that compressed storage is beneficial for every small matrix.



Comparison

`IsSparseMatrix()`

- ✓ Evaluates the number of zero elements relative to total matrix size.
- ✓ Focuses on the density of non-zero data.
- ✓ Uses the threshold defined by the current exercise.

`IsZeroMatrix()`

- ✓ Requires every element to be exactly `0`.
- ✓ Represents the strongest possible zero-density condition.
- ✓ Every zero matrix satisfies the sparse condition used here.

✦✦ Summary

- ✓ Counts zero elements inside a matrix.
- ✓ Calculates the total number of matrix positions.
- ✓ Compares zero density against the selected sparsity threshold.
- ✓ Demonstrates how density rules can be used to classify matrices.

Thank you 😊